The Morris Worm Incident of 1988

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Abstract

The Morris worm was a virus/worm developed by Robert Morris that infected a large portion of the internet after its release onto an MIT computer in the fall of 1988. Robert Morris reported that he had innocent intentions in his release of the worm, however the consequences of the worm's release led to economic and institutional harm to many networks. This paper discusses the resulting impact of that incident and the lasting influence it had on the field of cybersecurity. The first section of the white paper discusses Morris' development of the worm, its actions, its replication process, as well as the malfunction that caused the resulting network disruption. The second section of the white paper discusses the institutional response to the worm such as how it was detected and neutralized. The second section also discusses how the worm could have been mitigated against and what the response by institutions was to prevent events such as the Morris worm incident from happening in the future. The last section of the white paper discusses the trial and sentence of Robert Morris, as well as the legal and ethical controversies surrounding it. It is undeniable that the worm developed by Robert Morris had an untold impact on the field of cybersecurity that we are still feeling today.

Keywords: Robert Morris, cybersecurity, Morris worm, CERT, CFAA

The Morris Worm Incident of 1988

The Morris worm incident of 1988 was one of the most influential cybersecurity disasters to happen within the field. Prior to the event, the internet was a mostly trusting network of connected government entities, universities, and research institutions. The impetus of the event was the desire of Robert Morris, the creator of the worm, to measure and estimate the total size of the internet (FBI, n.d.). Despite his seemingly innocent intent, Morris' worm went far beyond its creator's design and caused a good deal of distress and destruction across the web. A feature within the worm had unintended consequences, causing it to repeatedly infect computers already possessing the worm, causing mass depletion of institutional resources (Spafford, 1988). The destructive aftermath of the worm eventually led to the conviction of Robert Morris under the CFAA, one of the first of its kind, establishing precedent for all subsequent CFAA cases (FBI, n.d.). Additionally, the poor communication surrounding the event led to the formation of the Computer Emergency Response Team. Lastly, the ethical boundaries crossed by Morris' experiment led many professional agencies to update their codes of ethics and conduct (Spafford, 1988). While the experimental Morris worm caused nationwide chaos and disruption, it also transformed and revolutionized cybersecurity by prompting technical, institutional, and ethical advancements.

In the fall of 1988, the Morris cyber worm was released onto the internet from a MIT computer, an incident that would have untold impact on the field of cybersecurity. The individual who released the worm, Robert Morris, was a twenty-three year old graduate student at Cornell, known for his Unix proficiency as well as his mischievousness (FBI, n.d.). Morris' intentions with the worm were reported not to be malicious, in his own words he was merely

attempting to “figure out the size of the internet” (Spafford, 1988). However, Morris’ worm ended up grinding many institutional and government networks to a standstill.

As stated before, the reported intention of the worm by Morris was an experiment to plumb the depths of the internet. The worm accomplished this by infiltrating Unix machines through shared networks, user accounts, files, as well as even randomly generating addresses (Spafford, 1988). The worm was incredibly successful at infiltrating due to taking advantage of many vulnerabilities in Unix systems at the time such as a buffer overflow in the “fingerd” service and the debugging feature in “sendmail” (Eichen & Rochlis, 1989). Once the malicious program had successfully infected a machine and scanned the network, the worm sent out “grappling hooks” to the systems it located which would then download the worm onto the new machine to start the whole process over again (Spafford, 1988). Morris’ idea was that for every fifteen worms deployed, a message would be sent back to one of his machines, the “size” of the internet could then be extrapolated by multiplying the number of messages he received by fifteen (Eichen & Rochlis, 1989).

If Morris’ intention was merely to determine the size of the internet, what exactly went wrong that shut down many important systems across the web? The function of the worm that ended up overwhelming many devices was an unintended consequence of one of the features of the program (Spafford, 1988). To improve the efficiency of the spread of the worm, Morris incorporated a switch where once a device was already infected, it would report to any potential new infections of the worm that the machine was already infected (Spafford, 1988). However, Morris knew that once others caught wind of his “experimentation”, a simple fix to repel the worm and end his project would be to falsely report that the machine had already been infected, almost like a vaccine. Anticipating this, Morris added an exception to this rule where one out of

seven times, the worm would ignore the “already infected” check and reinfect the machine. The “one out of seven” check proved to be entirely too frequent, it resulted in many systems being infected over and over repeatedly until resources were completely depleted.

MIT and Berkeley were two of the first institutions to defend against, capture, and dissect the worm. Berkeley was the first to identify that the worm was taking advantage of the “sendmail” and “fingerd” functions and distributed fixes for those to other institutions. Spafford (1988) credits Berkeley with closing most if not all the known vulnerabilities caused by the worm. MIT took extra care to capture the worm as well since the worm was introduced off of one of their own computers (Eichin & Rochlis, 1989). Besides the quarantining of the worm and patches provided by Berkeley, institutions also addressed the worm by isolating their network to prevent reinfection, creating worm detection scripts, and terminating any suspicious processes (Eichin & Rochlis, 1989).

While some of the lessons of the Morris worm incident could have only been learned through hindsight, there were many aspects of the incident that could have been prevented through proper mitigation. First, many institutions of that time had networks that were very permeable and had a high number of interconnected devices with little internal security, making it very easy for a worm to wreak havoc once it is on the inside (Eichen & Rochlis, 1989). Next, many of the weaknesses that the Morris worm took advantage of were already known to be weaknesses, such as the “rsh” and “rexec” host relationships and the “sendmail” debug mode (Spafford, 1988). These coupled with the close interconnectivity of networks made it incredibly easy for the Morris worm to infect and multiply across networks. Had these low security features been patched to begin with the worm would have been a lot less successful.

The Morris worm incident led to the disturbing realization that there was little to no response plan or way to communicate to deal with such a disaster. The GAO report (1989) discussing the incident mentions several security problems stemming from a lack of centralized internet security such as users not knowing how or who to report the virus to, difficulties in communicating information about the virus to vulnerable sites, confusion in who was responsible for protecting networks, etc. This confusion and lack of centralization is what lead to DARPA's funding and creation of the Computer Emergency Response Team, or "CERT" (GAO, 1989). The idea behind CERT was not only for it to be the central hub of coordinating responses to attacks, but also as the provider of information about known network vulnerabilities as well as the facilitator of security awareness and cooperation between institutions (GAO, 1989).

In 1990, Robert Morris was charged with violating the Computer Fraud and Abuse Act of 1986, or CFAA (United States v. Morris, 1990). To back this up, the federal grand jury stated that Morris had accessed federal computers without authorization, caused at least $1,000 in damages, and altered, damaged and prevented authorized use of computers to others (United States v. Morris, 1990). This was corroborated by the fact that he released the worm on a computer he had no authorization to use (MIT's network), and the fact that it reached systems as far as NASA's. This conviction made Robert Morris one of the first to be prosecuted for violating the CFAA (FBI, n.d.). Despite this, Morris was only sentenced to three years' probation, a $10,000 fine, and four hundred hours of community service (FBI, n.d.).

The legal and ethical issues that arose from the trial was the controversy around whether violation of the CFAA required proof that Robert Morris had prior intent to cause the damage that resulted from his worm (Chen, 1990). The contentious passage in question from the CFAA states "intentionally accesses a Federal interest computer without authorization, and by means of

that conduct causes damage. . .” (United States v. Morris, 1990). Morris argued that his programming failure is what caused the unauthorized access as well as the damaged caused by crashes and loss of availability and that there was no intention involved (United States v Morris, 1990). Despite this, the court argued that the worm Morris created was designed intentionally to enter systems and computers where he did not have authorization, resulting in the infection of computers belonging to universities, the military, and NASA (Chen, 1990).

This debate had lasting influence on the cybersecurity field because it helped define the borders between responsible academic curiosity and irresponsible hacking. It emphasized that just because networks are accessible, it does not mean that you’ve been granted authorization to enter (Reynolds, 1989). Additionally, despite the fact that data and system integrity were not harmed, loss of availability in the form of wasted computation, staff time, are still financial damages (Chen, 1990). The debate also raised the standards of academic standards for network experimentation, requiring consent from both parties, authorization, and effective mechanisms for shutdown (Chen, 1990). In regard to development of ethical codes, shortly after the Morris worm incident, the ACM Code of Ethics, NSF, Computer Professionals for Social Responsibility, the National Science Foundation, and the Internet Activities Board all published ethical statements or revisions to existing ethical codes (Chen, 1990).

While the experimental Morris worm caused nationwide chaos and disruption, it also transformed and revolutionized cybersecurity by prompting technical, institutional, and ethical advancements. Morris took advantage of known security flaws within Unix systems, but he also took advantage of the institutional trust that existed at the time, trust that has never fully come back. It has only led to more security protocols and policies that have likely hindered future research and cooperation between institutions. While there are many things that hinder

advancements in cybersecurity such as bureaucracy and budget constraints, many companies and institutions are very resistant to even trusted cybersecurity professionals finding security flaws in their systems. Whether it is because of short term financial shortsightedness, or distrust, it is undeniable that the Morris worm incident has had ripples that have had influence event today.

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